

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Floating Point Representation

Lecture No.- 2

By- Vishvadeep Gothi sir



Recap of Previous Lecture



Topic

Floating Point Representation



Topics to be Covered



Topic

Floating Point Representation

Topic

IEEE Floating Point Representation

$$e = E - \text{bias}$$

#Q. A certain well-known computer family represents the exponents of its floating-point numbers as "excess-64" integers; i.e., a typical exponent $e_6e_5e_4e_3e_2e_1e_0$ represents the number:

$\rightarrow$ bias = 64
 $\rightarrow$ binary $\Rightarrow$ decimal - 64
 E

A $e = -64 + \sum_{i=0}^6 2^i e_i$

B $e = -64 + \sum_{i=0}^6 2^i e_i$

C $e = 64 - \sum_{i=0}^6 2^i e_i$

D $e = 64 - \sum_{i=0}^6 2^i e_i$

$$\left(e_0 2^0 + e_1 2^1 + e_2 2^2 + \dots + e_6 2^6 \right) = 64$$

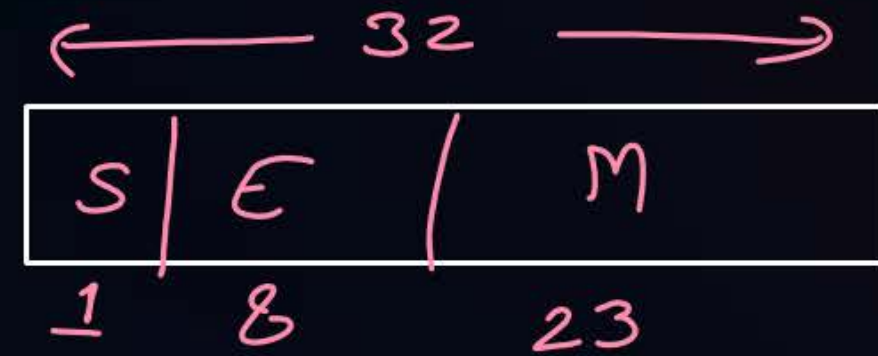
$$\sum_{i=0}^6 e_i 2^i = 64$$



Topic : IEEE-754 Floating Point Representation

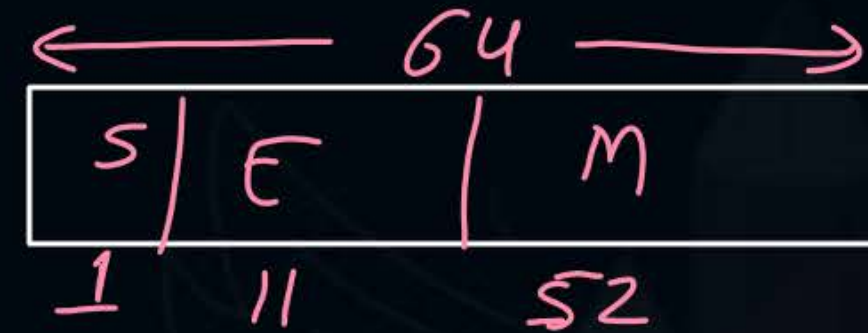
IEEE-754 Representation

Single Precision



bias = 127

Double Precision



bias = 1023

$E = 00\dots0$
or
 $11\dots1$ } Special

$E = 0000\dots01$
to
 $1111\dots10$ }

normal
number
(Implicit
normalized)



Topic : IEEE-754 Floating Point Representation

S	E	M	Number
0	00.....0	00.....0	+0
1	00.....0	00.....0	-0
0	11.....1	00.....0	+∞
1	11.....1	00.....0	-∞
0 or 1	11.....1	$M \neq 00.....0$	N.A.N. (Not A Number)
0 or 1	000.....0	$M \neq 00.....0$	Denormalized
0 or 1	$E \neq 00.....0$ and $E \neq 11.....1$	00.....0 to 11.....1	Normal number (Implicit normalized)

N.A.N. :-

for single precision $\Rightarrow$ bias = 127

$E \Rightarrow$ represented by = $\frac{8 \text{ bits}}{\Downarrow}$

range of values possible for $e \Rightarrow -128$ to $+127$

original exponent $E_{\max} = +127$

after biasing

$$E_{\max} = +127 + 127$$

$$= (254)_{10}$$

$$= (11111110)_2$$

} hence $E = (111\dots1)_2$ not possible for any number.



Topic : Denormalized Number

very small numbers which can not be stored as normalized.

For IEEE-754 single precision:-

$$\text{bias} = 127$$

for Normalized number $E_{\min} = (00000001)_2 = (1)_{10}$

$$\begin{aligned} e_{\min} &= E_{\min} - \text{bias} \\ &= 1 - 127 \\ &= -126 \end{aligned}$$

ex 1 :-

0. 00000000000000000101
125 zero

↓
implicit normalizedⁿ

↓
 $1.01 * 2^{-126}$

$m = 01$

$e = -126$

$E = -126 + 127 = 1$
 $= (00000001)_2$

ex 2 :-

0. 00000000000000000101
126 zero^s

↓
normalize

↓
 $0.101 * 2^{-126}$

↓
store the number as denormalized number

	E	M
S	00.....0	10100...0



Topic : Value Formula



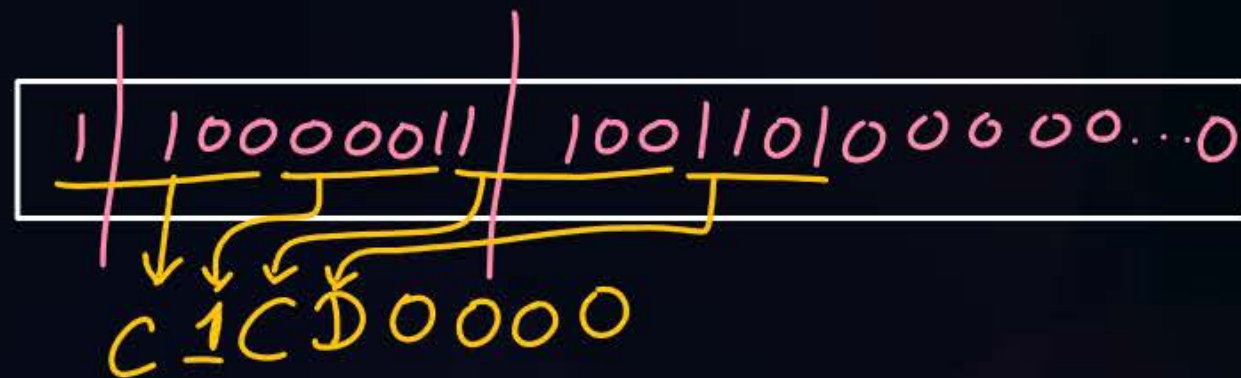
for double precision
↓

-1022

$$\text{Value (denormalized)} = (-1)^S * 0.M * 2^{-126} \quad \text{or}$$

$$\text{Value (normalized)} = (-1)^S * 1.M * 2^{E - \text{bias}}$$

#Q. The value of a float type variable is represented using the single-precision 32-bit floating point format IEEE-754 standard that uses 1 bit for sign, 8 bits for biased exponent and 23 bits for mantissa. A float type variable X is assigned the decimal value of -25.625 . The representation of X in hexadecimal notation is?



$$S = 1$$

$$(11001.101)_2$$

↓
implicit normalize

$$1.1001101 \times 2^4$$

$$e = 4$$

$$E = 4 + 127 = (131)_{10} = (10000011)_2$$

$$m = 10011010000000000000000$$

[NAT]

Ans = +26



#Q. The value represented by the following 32-bits in IEEE-754 representation is?

0100000111010000...00

S E M

$E \neq 00\dots0$
and
 $E \neq 11\dots1$ } $\Rightarrow$ Implicit normalized.

$E = (10000011)_2$
 $\Downarrow$
 $= (131)_{10}$

value = $+ 1.101000\dots0 \times 2^{131-127}$
 $= 1.10100\dots0 \times 2^4$
 $= + (11010.0000)_2$
 $= + (26)_{10}$

[NAT]



#Q. The value represented by the following 32-bits in IEEE-754 representation is?

00000000001000000...00
E m

$E = 00 \dots 0$
 $m \neq 00 \dots 0$ } denormalized

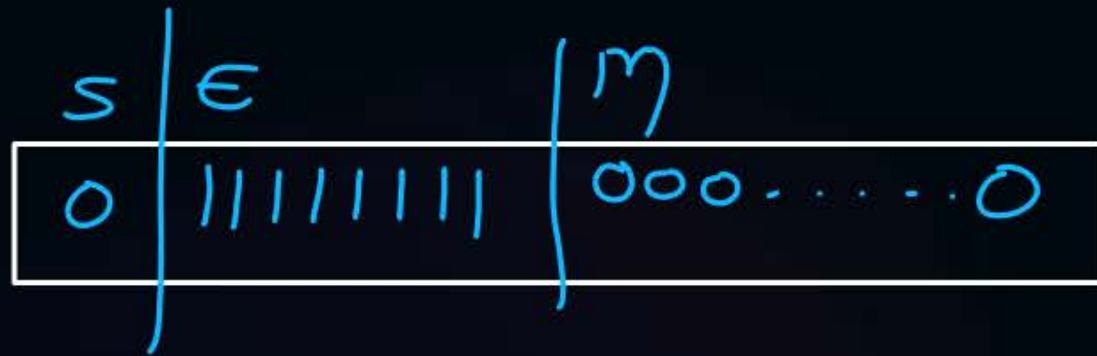
$$\begin{aligned} \text{value} &= + 0.101 * 2^{-126} \Rightarrow 0.625 * 2^{-126} \\ &= + (101.0) * 2^{-3} * 2^{-126} \\ &= 5 * 2^{-129} \end{aligned}$$

[NAT]



#Q. Maximum value represented in IEEE-754 single precision?

$+\infty$



[NAT]



#Q. Minimum value represented in IEEE-754 single precision?

$-\infty$

S	E	M
1	11111111	00000...0

[NAT]



#Q. Maximum positive normalized value represented in IEEE-754 single precision?



$$\begin{aligned}\text{Value} &= + 1.111\dots 1 * 2^{254-127} \\ &= + 1111\dots 1.0 * 2^{-23} * 2^{127} \\ &= + (2^{24}-1) * 2^{104}\end{aligned}$$

[NAT]



#Q. Minimum normalized value represented in IEEE-754 single precision?

s	E	M
1	111...10	111...1

$$\text{Value} = - (2^{24} - 1) 2^{104}$$

[NAT]



#Q. Minimum positive normalized value represented in IEEE-754 single precision?

S	E	M
0	00000001	000-----0

$$\begin{aligned}\text{Value} &= + 1.00000\dots 0 * 2^{1-127} \\ &= + 2^{-126}\end{aligned}$$

[NAT]



#Q. Maximum negative normalized value represented in IEEE-754 single precision?

S	E.	M
1	00000001	00...0

$$\text{Value} = -2^{-126}$$

[NAT]



#Q. Maximum positive denormalized value represented in IEEE-754 single precision?

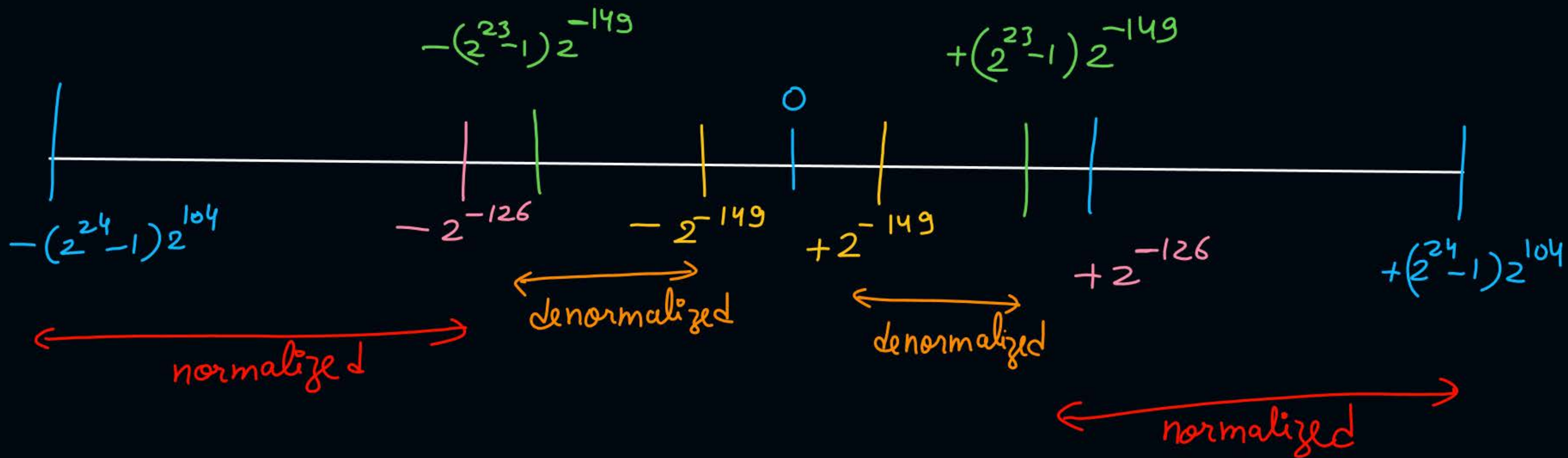
S	E	M
0	00000000	1111.....1

$$\begin{aligned}\text{value} &= + 0.111\dots 1 * 2^{-126} \\ &= + 1111\dots 1 * 2^{-23} * 2^{-126} \\ &= + (2^{23} - 1) * 2^{-149}\end{aligned}$$

#Q. Minimum positive denormalized value represented in IEEE-754 single precision?



$$\begin{aligned}
 \text{value} &= + 0.000000\dots01 * 2^{-126} \\
 &= +(1.0) * 2^{-23} * 2^{-126} \\
 &= + 2^{-149}
 \end{aligned}$$



[NAT]



#Q. How to represent $-(1)_{10}$ in IEEE-754 single precision floating point number?

$$\begin{array}{c} (1)_{10} \\ \Downarrow \\ -(1.0)_{10} = -(1.0)_2 \Rightarrow 1.0 * 2^0 \end{array}$$

S	E	M
1	01111111	000...0

$$m = 000 \dots 0$$

$$e = 0$$

$$\begin{aligned} E &= 0 + 127 = 127 \\ &= (01111111)_2 \end{aligned}$$

#Q. How to represent $+(0.0000101)_2$ in IEEE-754 single precision floating point number?

$s = 0$

$\Downarrow$
 implicit normalization
 $\Downarrow$
 $1.01 * 2^{-5}$

$m = 0100 \dots 0$

$e = -5$

$E = -5 + 127 = 122$

S	E	M
0	01111010	0100.....0

#Q. The value of a float type variable is represented using the single-precision 32-bit floating point format IEEE-754 standard that uses 1bit for sign, 8 bits for biased exponent and 23 bits for mantissa. A float type variable X is assigned the decimal value of -14.25 . The representation of X in hexadecimal notation is

A

C1640000H

B

416C0000H

C

41640000H

D

C16C0000H

[NAT]



H.W.

#Q. Consider the following representation of a number in IEEE 754 single-precision floating point format with a bias of 127.

S: 1 E: 10000001 F: 111100000000000000000000

Here S, E and F denote the sign, exponent and fraction components of the floating-point representation.

The decimal value corresponding to the above representation (rounded to 2 decimal places) is_____



[NAT]

H. W.



#Q. The format of the single-precision floating-point representation of a real number as per the IEEE 754 standard is as follows:

Sign	Exponent	mantissa
------	----------	----------

Which one of the following choices is correct with respect to the smallest normalized positive number represented using the standard?

- A. exponent = 00000001 and mantissa = 00000000000000000000000000000001
- B. exponent = 00000001 and mantissa = 00000000000000000000000000000000
- C. exponent = 00000000 and mantissa = 00000000000000000000000000000000
- D. exponent = 00000000 and mantissa = 00000000000000000000000000000001



2 mins Summary



Topic

Floating Point Representation

Topic

IEEE Floating Point Representation





Happy Learning

THANK - YOU

